

MILESTONE 1

Revenue Leakage at Revo Hospital

An analytical dashboard for the finance lead: tracking IDR 14.5 billion of revenue still uncollected across the Java network

DATAVIZ_JUN26_Reinaldy_Zulfananda_Arkaan_intermediate

Data Visualisation Intermediate Assignment • RevoU Full Stack Data Analytics • September 2026



Dashboard Angle

MILESTONE 1

The single question this dashboard answers

Target user

Finance lead of the Revo Hospital network, the owner of collection and cash flow decisions across branches.

Decision

Which patient segments and periods to chase first so pending revenue falls below the 10 percent tolerance.

Why it matters

IDR 14.5 billion is locked up, equal to 14.7 percent of revenue. That cash cannot fund branch operations.

Dashboard type

Analytical. It is built to find causes and patterns, not to monitor daily activity.

Why analytical rather than strategic or operational

- A strategic dashboard needs long-horizon targets and market benchmarks, neither of which exists in this dataset.
- An operational dashboard needs daily data and immediate action, while the finest grain here is an admission with monthly patterns.
- What the finance lead needs is to trace where receivables pile up, so the dashboard works as a diagnosis tool.

Data Preparation

Five tables, one fact table, grain of one admission per row

Table	Key	Why it is necessary
admissions	admission_id	The fact table. It sets the grain of the analysis and holds the admission date.
billing	admission_id	Source of net revenue, out of pocket, and payment_status. The core of the angle.
patients	patient_id	Supplies insurance_type, the main comparison dimension.
branches	branch_id	Branch identity and coordinates for geographic analysis.
departments	department_id	Service line context when tracing the source of receivables.

Relationship direction

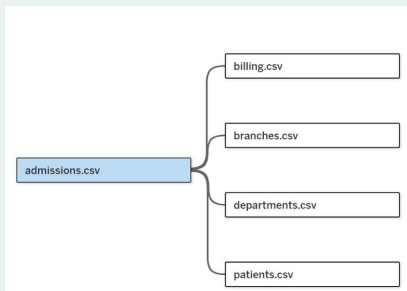
billing 1 : 1 admissions

patients 1 : many admissions

branches 1 : many admissions

departments 1 : many admissions

Data Model



Risk if the join is wrong

If billing were joined to patients without passing through admissions, each bill would duplicate once per visit. Total revenue would inflate and the pending rate would be wrong. The Grain Validation sheet confirms that the row count matches the count of distinct admission_id.

Metrics & Logic

Five metrics, all pointing at the same question

Metric	Type	Why it is useful to the finance lead
Total Net Revenue	Aggregate	The denominator behind every ratio and the measure of network scale.
Uncollected Revenue	Aggregate	The rupiah value sitting idle. This is the number that gets acted on.
Pending Revenue Rate	Ratio	Compares branches and segments fairly regardless of their size.
Out of Pocket Ratio	Ratio	Tests whether a heavier patient burden is what drives bills into pending.
Revenue per Admission	Ratio	Separates revenue growth from volume out of growth from higher bill value.

Ratio requirement

Two metrics are ratios: Pending Revenue Rate and Out of Pocket Ratio.

Time-based requirement

Admission Month, built with DATETRUNC, splits Pending Revenue Rate into 24 monthly points.

KPI Scorecard

Four numbers that size the problem

IDR 99B

billion in net revenue over
two years

IDR 15B

billion in revenue still
uncollected

15%

pending revenue rate

IDR 20M

million revenue per admission

Why a scorecard

The four numbers build context in layers. Revenue shows scale, uncollected shows the value at risk, the pending rate shows how severe that is in relative terms, and revenue per admission gives a reference point when the rate moves. Each card title is written as a finding, not as a metric name.

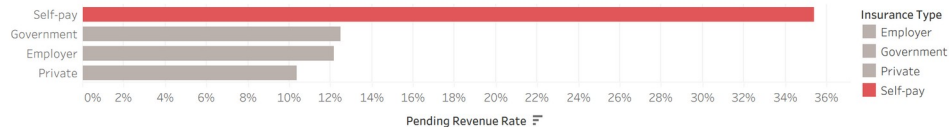
Weaker alternative

A summary table holding all four figures at once. Every value would carry equal weight, so the reader would not know which one is urgent, and the IDR 15 billion would lose its emphasis.

Comparison Chart

Self-pay patients leave 35% unpaid, three times any other segment

Self-pay patients leave 35% unpaid, three times any other segment



Pending Revenue Rate for each Insurance Type. Color shows details about Insurance Type.

Purpose

Identify which patient segment leaves the most bills sitting in pending status.

Why a bar chart

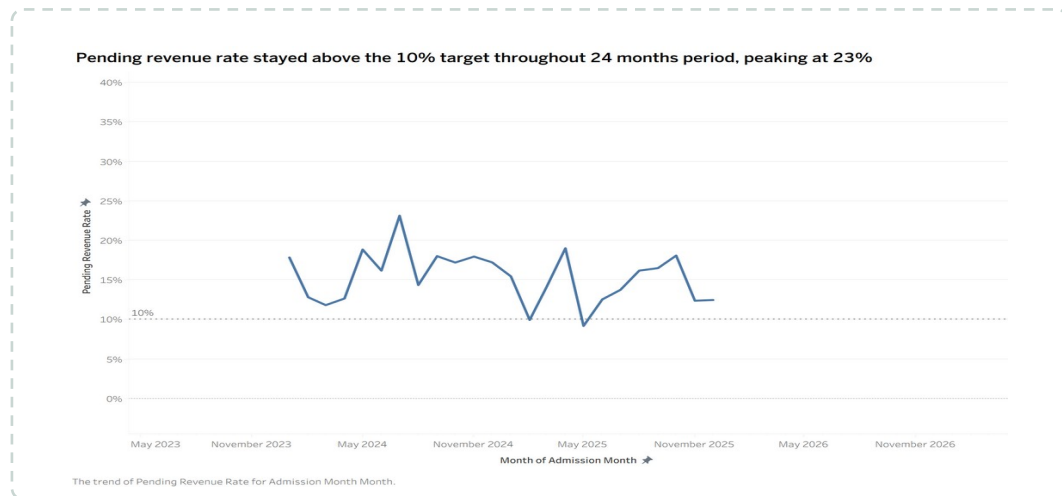
Comparing bar length is the easiest read for a handful of unordered categories.

Weaker alternative

A pie chart. Pending rate is a ratio within each segment, not a share of one whole, so the slices would not add to 100 percent and would mislead. The gap between 35 percent and roughly 12 percent is also harder to judge by angle than by length.

Trend Chart

Pending revenue rate stayed above the 10% target in 22 of 24 months, peaking at 23%



Purpose

Test whether the collection problem improves on its own or persists across time.

Why a line chart

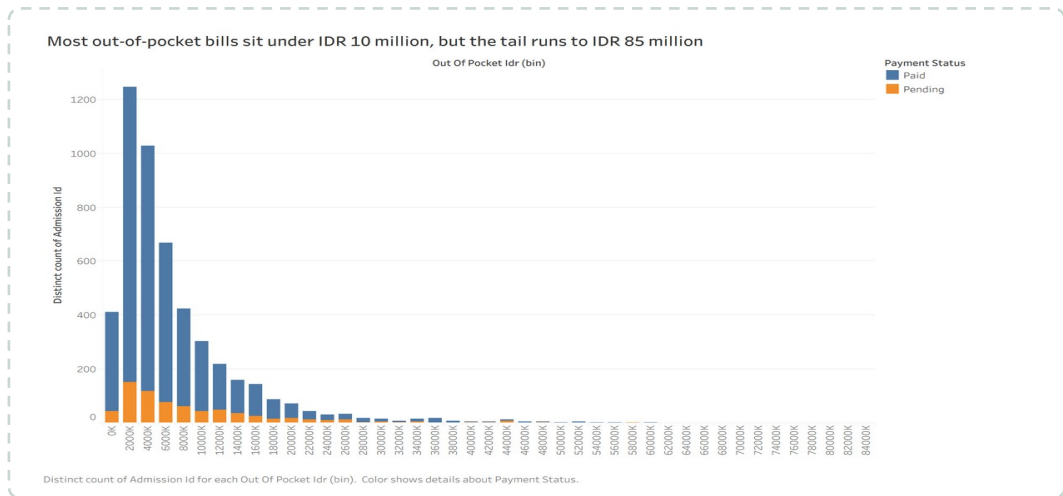
A line carries continuity between months, and the 10 percent reference line makes the distance to target visible at once.

Weaker alternative

A monthly bar chart. Bars break the sense of flowing time and push the reader to compare months one by one, when the question is the direction over two years. A monthly percent difference was also tried and dropped, because one month with a small denominator spiked to 287 percent.

Distribution Chart

Most out-of-pocket bills sit under IDR 10 million, but the tail runs to IDR 85 million



Purpose

See how patient burden is spread and find the tail of large bills most at risk of going unpaid.

Why a histogram

A two million rupiah bin reveals the shape of the spread and the length of its tail, which an average hides.

Weaker alternative

A single average out-of-pocket KPI. One number conceals the tail reaching IDR 85 million, and those large bills are exactly the ones most likely to become receivables. A box plot is more compact but harder to read for a non-technical audience such as a finance lead.

Milestone 1 Status

Every mandatory building block is now in place in Tableau

5 tables

Above the minimum of 4

5 metrics

Including two ratios and one monthly view

Comparison

Pending rate by insurance type, sorted

Distribution

Out-of-pocket bins of two million rupiah

Relationships & grain

Admissions at the centre, one row per admission

4 KPI cards

Above the minimum of 3

Trend

24 monthly points with a 10 percent reference line

Chart titles

All written as findings rather than chart names

Milestone 2 Technical Build

Five advanced techniques layered on top of the Milestone 1 building blocks

LOD expression

Two-year branch revenue that stays the same when filters change

Top 5 ranking

Departments ordered by pending revenue rate

Running total

Pending revenue added up by admission month

Branch map

Eight branches plotted from latitude and longitude

Three filters

Admission date, branch name, insurance type

Seven insights

OBIPR findings close the deck

Grain unchanged

Still one row per admission across every new view

Published

[public.tableau.com/views/
CollectionRiskDashboardRevoHospitalJavaNetwork](https://public.tableau.com/views/CollectionRiskDashboardRevoHospitalJavaNetwork)

Level of Detail

Branch totals that stay fixed while the rest of the dashboard filters



Purpose

Keep each branch's full two-year revenue as a reference while the user filters a single month or segment.

Why the number stays fixed

It counts every month and segment, so filters do not move it. The tooltip labels it as unfiltered.

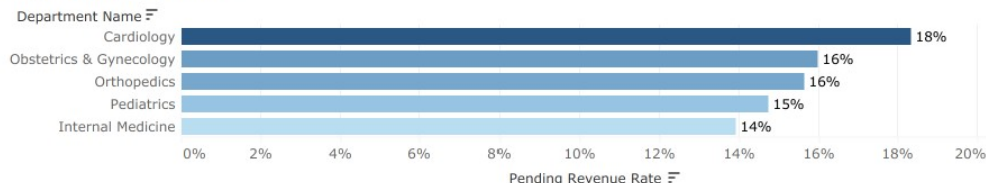
Weaker alternative

A plain SUM of net revenue. It shrinks with every filter, so the user loses the denominator and can no longer tell whether the filtered slice is a large or a small part of that branch. A table calculation would break in the same way, because it recomputes on whatever rows the filter leaves behind.

Ranking

Cardiology tops the pending rate at 18.3%, more than three points above the network average

Top 5 Departments



Purpose

Narrow eight departments down to the five where receivables build up fastest.

Why a sorted bar

A Top 5 filter with a descending sort makes the order itself the message.

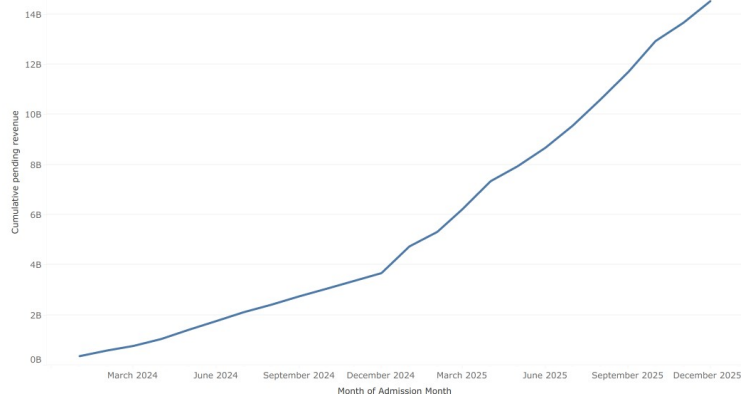
Weaker alternative

Showing all eight departments. The spread runs from 13 to 18 percent, so the lower half adds length without adding a decision. Ranking by rupiah instead of rate was also tested: it puts Oncology first at IDR 3 billion, which answers a different question about where the largest amounts sit.

Table Calculation

Pending revenue summed by admission month reaches IDR 14.5 billion, climbing faster from 2025

Pending revenue added up by admission month reaches IDR 14.5B



Purpose

Show which admission months created the bills that are still pending, added up over time.

What it is and is not

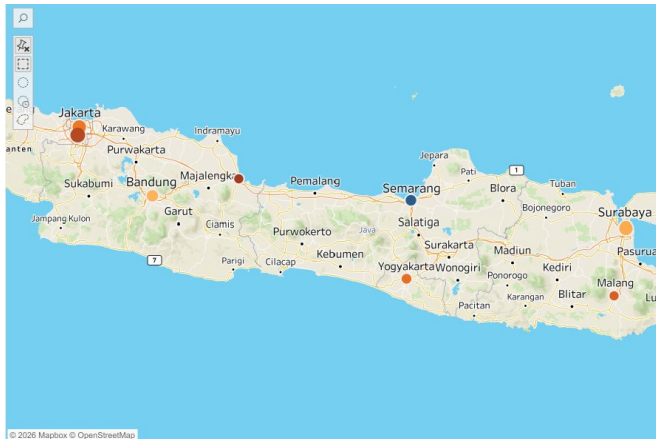
A cumulative sum by admission month. Not a month-end receivable balance, the data has no payment date.

Weaker alternative

Monthly pending revenue on its own. Each bar looks small, and the reader never sees that bills admitted in 2024 already make up about IDR 4 billion of what is pending today, before the 2025 volume surge tripled the monthly intake.

Map Chart

Geography is not the problem: every branch sits between 12% and 16% pending



Purpose

Test whether the collection problem is concentrated in one city or spread across the Java network.

Why a symbol map

Colour carries the pending rate and size carries branch revenue, so scale and risk read in one glance.

Weaker alternative

A filled map shaded by province. Branches are points, not territories, so shading a whole province would credit a rate to areas the hospital does not serve. The flat spread is the finding here, and it moves the fix from branch management to network-wide payment policy.

Dashboard Interaction

Three filters, one of them a date range, applied across every worksheet

Filter	Type	What it lets the finance lead do
Admission Date	Range of dates	Isolate a quarter or the 2025 surge and see whether the pending rate moved with it.
Branch Name	Multiple values	Check one branch against the network band of 12 to 16 percent.
Insurance Type	Multiple values	Remove Self-pay and confirm the remaining segments sit near 12 percent.

Applied to all worksheets

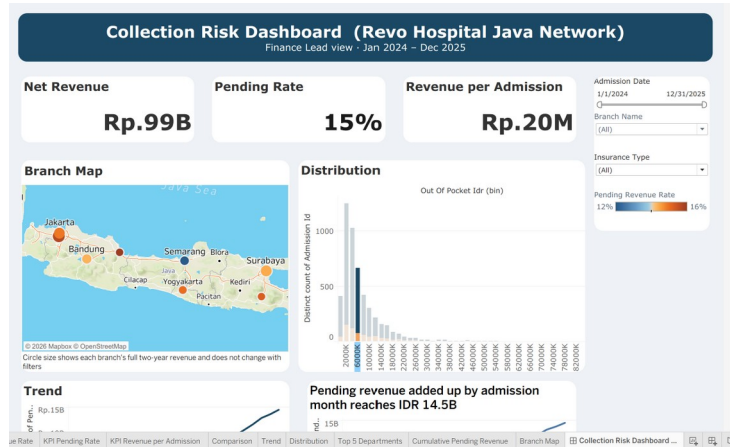
Every filter uses All Using This Data Source, so one selection moves the KPI cards, the map and all four charts together.

Reading order

KPI cards first, then segment and geography, then the time and ranking views that explain where the backlog sits.

Final Dashboard

Both halves of the final dashboard, eight views driven by the same three filters



Layout hierarchy

Row one sizes the problem, row two locates it, row three explains it over time and by department.

Interaction flow

Filters sit in a fixed right rail, so they stay in reach while the lower rows are scrolled.

Trade-off accepted

The canvas is 1200 by 1100, so the lower rows need a short scroll. Compressing everything into one screen would have shrunk the map and the histogram below a readable size, and the three KPI cards plus the segment chart already carry the headline before any scrolling happens.

Insights

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Seven OBIPR findings with observation, business impact, priority and action

Insight	Observation	Business impact	Priority and action
Self-pay drives the gap	35% unpaid against 10 to 13% for insured segments	13% of admissions but 30% of all pending revenue, about IDR 4B	High. Deposit or payment plan for bills above IDR 20M
The backlog is systemic	All eight branches sit between 12% and 16% pending	No single branch fix recovers the money, the IDR 14.5B is spread across every site	High. Change payment policy at network level
Volume multiplies exposure	Monthly revenue tripled from IDR 2B to IDR 6B	New pending revenue grew from about IDR 300M to IDR 900M a month	Medium. Scale the collections team with volume
Cardiology and Oncology lead	Cardiology has the highest rate at 18%	The two departments hold about IDR 6B, 40% of all pending revenue	High. Structured instalments in both departments
Large bills fail more often	Median bill IDR 6M, tail up to IDR 85M	Bills above IDR 20M are 6% of cases but 23% of pending revenue	High. Financial counselling above IDR 20M
Patients carry too much	39% of gross charges fall on the patient	About IDR 40B of charges depend on household ability to pay	Medium. Widen insurer partnerships
It never resolved itself	Above the 10% target in 22 of 24 months, peak 23%	At the 10% target the backlog would be IDR 10B, so about IDR 5B is excess	High. Monthly review against the 10% target

Closing Reflection

What worked, what was traded away, and what the next version would change

What worked

One angle held across all eight views, so no chart is decorative

Trade-off

The distribution keeps its long tail, so the axis stays wide

Next version

Add a department by segment cross view to pinpoint the worst cell

Next version

Publish a narrower layout for the same eight views

What worked

Chart titles written as findings, which shortened the walkthrough

Trade-off

The map earns little space, but it proves the risk is not geographic

Next version

Add an ageing bucket once a payment date column is available

Open question

Whether a deposit rule would also reduce Self-pay admissions

END OF REPORT

Thank you

Reinaldy Zulfananda Arkaan • September 17, 2026

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Linkedin : <https://www.linkedin.com/in/reinaldy-zulfananda-arkaan-30567a179/>

Live dashboard on Tableau Public

https://public.tableau.com/views/CollectionRiskDashboardRevoHospitalJavaNetwork_17896636711850/

CollectionRiskDashboardRevoHospitalJavaNetwork?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link